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Forecasting seasonal demand for retail: A Fourier time-varying grey model

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ABSTRACT

Seasonal demand forecasting is critical for effective supply chain management. However, conventional forecasting methods face difficulties accurately estimating seasonal variations, owing to time-varying demand trends and limited data availability. In this paper, we propose a Fourier time-varying grey model (FTGM) to tackle this issue. The FTGM builds upon grey models, which are effective with limited data, and leverages Fourier functions to approximate time-varying parameters that allow it to represent seasonal variations. A data-driven selection algorithm adaptively determines the appropriate Fourier order of the FTGM without prior knowledge of data characteristics. Using the well-known M5 competition data, we compare our model with state-of-the-art forecasting methods taken from grey models, statistical methods, and architectures of neural network-based methods. The experimental results show that the FTGM outperforms popular seasonal forecasting methods in terms of standard accuracy metrics, providing a competitive alternative for seasonal demand forecasting in retail companies.

1. Introduction

Demand forecasting is the basis of most decisions for supply chain management. Demand volatility hugely impacts retail networks and their ability to perform in an efficient yet responsive way (Falatouri, Darbanian, Brandtner, & Udokwu, 2022). Forecasting the future demand of a retail company plays a pivotal role in marketing, sales, purchasing, pricing, ordering, and inventory management (Hoeltgebaum, Borenstein, Fernandes, & Veiga, 2021). As pointed out by Fildes, Ma, and Kolassa (2022), high-accuracy demand forecasting has a substantial impact on organizational performance because it improves many processes along the retail supply chain. Thus, ensuring accuracy in demand forecasting is important, given that overstocking products results in additional costs while understocking leads to lost sales and diminished customer satisfaction.

Seasonality is a prevalent characteristic in retail demand data (Ehrenthal, Honhon, & Van Woensel, 2014; Guo, Fang, Zhao, & Wang, 2021). Forecasting seasonal demand in the retail sector is notably challenging, primarily due to the limitation of short data histories. Retail industries often have a small sample size of available data, either because their products are relatively new to the market or because older data are discarded in the belief that they are not relevant to the forecasting task (Svetunkov & Boylan, 2020). Consequently, even when products, outlets, and ordering modes are well in place, historical records may not be retained long enough to accurately capture seasonality (Svetunkov, Chen, & Boylan, 2023). Therefore, the development of models capable of forecasting seasonal demand with limited complete seasonal cycles of data holds great significance in the retail sector.

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Numerous forecasting methods have been utilized for seasonal demand, such as exponential smoothing state-space models, autoregressive integrated moving average models, Fourier terms, seasonal dummy variable-based methods, and neural networks (De Livera, Hyndman, & Snyder, 2011; Keller, Deleersnyder, & Gedenk, 2019; Petropoulos et al., 2022; Wolters & Huchzermeier, 2021). However, most of these methods face challenges when dealing with seasonal forecasts in retail with limited data samples. Further, the parameters of some models remain constant over time, failing to account for the dynamic nature of model structures in practice. For example, the effects of marketing activities on product sales may vary over time (Huang, Fildes, & Soopramanien, 2019). Similarly, changes in customer preferences or lifestyles can influence fluctuations in demand Meeran, Jahanbin, Goodwin, and Quariguasi Frota Neto (2017). Also, the entry of new competitors can reduce the effect of prices and promotions on the demand for the focal product. As a result, models with invariant parameters often fall short of capturing these dynamic changes.

As noted by Liu, Yang, and Forrest (2017), the grey model is effective at handling forecasting problems on small sample data. Some scholars have incorporated seasonal elements into grey models for forecasting seasonal time series, such as seasonal factors, dummy variables, and trigonometric functions (Comert, Begashaw, & Huynh, 2021; Wang, Li, & Pei, 2018; Zhou & Ding, 2021). However, these models are not adaptive for complex multi-frequency seasonal patterns. Although (Wang, Xie, & Yang, 2022) introduced the Fourier function to represent seasonal forcing, it fails to capture time-varying growth rates, since it uses a fixed development coefficient.

In this paper, we propose a Fourier time-varying grey model (FTGM) for seasonal retail demand forecasting with small sample data. Unlike traditional grey models, the FTGM employs a reduced-order integro-differential equation to directly model the original time series. The proposed model is evaluated using the M5 competition datasets, and a rolling origin evaluation strategy is used for model comparison. The main contributions of our paper are summarized as follows:

- We propose the FTGM by leveraging Fourier series to approximate time-varying parameters for both the development coefficient and the forcing term. The model effectively captures complex seasonal variations in demand data.
- We introduce integral matching for parameter estimation and present a data-driven selection algorithm that adaptively determines the appropriate Fourier order of the FTGM.
- We conduct extensive comparative studies between the FTGM and 11 state-of-the-art forecasting methods on the publicly available M5 competition datasets using an out-of-sample rolling origin evaluation strategy.

The remainder of this paper is organized as follows. Section 2 reviews related studies on seasonal demand forecasting. Section 3 introduces our Fourier time-varying grey model, describes the utilization of integral matching for parameter estimation, and outlines our order selection algorithm. Section 4 describes the experimental design, encompassing data, benchmark models, error metrics, and the procedure of rolling origin evaluation and model training. Section 5 presents the empirical results and discussions, which include accuracy and ranking comparisons, statistical testing analysis, ablation experiment analysis, and an assessment of the model's performance on shorter series. Finally, Section 6 concludes our work and outlines potential areas for future research.

2. Related work

Seasonal time series forecasting has been a recurring research topic in management science. Traditional approaches for such forecasting involve statistical methods such as exponential smoothing (ETS) and the autoregressive integrated moving average (ARIMA). Seasonal ARIMA uses seasonal differences to remove non-stationarity from time series data (Box, Jenkins, Reinsel, & Ljung, 2015). The Holt–Winters model supplements Holt's linear trend model with an additive or multiplicative seasonal component, suitable for forecasting data with both trend and seasonal variation (Winters, 1960). A taxonomy of ETS methods has been proposed to account for diverse trend and seasonal component combinations (Hyndman, Koehler, Ord, & Snyder, 2008). Some of these methods are straightforward to implement but may suffer from severe parameterization. Recent studies (Svetunkov & Boylan, 2020; Svetunkov et al., 2023) have raised concerns about the utility of seasonal ARIMA and seasonal ETS models in retail contexts because of numerous parameters to estimate. Particularly in scenarios with short data histories, accurately estimating these parameters and maintaining their stability can pose challenges. In addition, Dekker, van Donselaar, and Ouwehand (2004) found that the Holt–Winters model failed to accurately estimate seasonality in the presence of high demand uncertainty and stochastic seasons.

Another important set of methods uses Fourier terms (or trigonometric formulation) in statistical models for forecasting time series with seasonal patterns. The Box–Cox, ARMA, trend, seasonal (BATS) model was developed to address the challenges associated with seasonal time series, including modeling multiple seasonal periods and non-integer seasonality (De Livera et al., 2011). However, the BATS model may not perform well when the seasonality is complex and high-frequency. Trigonometric BATS (TBATS) and dynamic harmonic regression ARIMA (DHR-ARIMA) are advocated as more suitable methods for forecasting seasonal time series than ETS and ARIMA (Godahewa, Bergmeir, Webb, & Montero-Manso, 2023). TBATS employs a Fourier-series-based trigonometric representation to model seasonality, making it one of the state-of-the-art methods for forecasting seasonal time series (De Livera et al., 2011). DHR-ARIMA leverages Fourier terms and ARMA errors to capture seasonality and time series dynamics, and is frequently applied in the context of supply chain forecasting (Hyndman & Athanasopoulos, 2021). For example, Wolters and Huchzermeier (2021) used a harmonic regression model to predict seasonal sales for a grocery retailing chain. These methods are effective at forecasting long seasonal cycles.

The decomposition approach is also a popular strategy used by statistical methodologies to handle seasonality. This method involves decomposing a time series into three components: trend, seasonal, and residual (Ord, Fildes, & Kourentzes, 2017). The de-seasonalized time series is predicted using a non-seasonal method, and then the seasonal component is added back to the predictions. For instance, Theodosiou (2011) employed seasonal–trend decomposition using the LOESS (STL) procedure to decompose the components of time series and applied linear combinations of the disaggregated sub-series for forecasting in the NN3 and M1 competition series. Xie, Qian, and Wang (2020) developed a complete ensemble empirical mode decomposition with adaptive noise for forecasting tourism demand. However, the accuracy of this approach depends heavily on the choice of the decomposition technique and the method of combination used (additive, multiplicative, or other mixed methods).

More recently, machine learning algorithms have found applications in seasonal time series forecasting. Compared with the statistical models described above, the main advantage of these approaches is that they eliminate the need for linearity assumptions or specific assumptions about the distribution of residuals. Among these competitive methods, neural networks (NNs) have been advocated as strong alternatives for seasonal time series forecasting (Zhang, Eddy Patuwo, & Y. Hu, 1998). In particular, recurrent NNs (RNNs) and convolutional NNs (CNNs) have exhibited promising results, outperforming many state-of-the-art statistical forecasting techniques (Bandara, Bergmeir, & Smyl, 2020). The long short-term memory (LSTM) technique, a type of RNN, adeptly handles diverse seasonal periods within time series data (Bandara, Bergmeir, & Hewamalage, 2021). Abbasimehr, Shabani, and Yousefi (2020) proposed a demand forecasting method based on multilayer LSTM networks. However, machine learning-based methods are prone to overfitting and local optimization problems, which may lead to serious fluctuations in forecasts, potentially falling short of expected accuracy levels.

Unlike the aforementioned time series forecasting models, the grey model proposed by Deng (1984) has shown advantages in handling small sample data. These advantages primarily arise from three aspects: (1) grey models do not require prior knowledge of data distribution, but rather employ the cumulative sum operator (or integral operator) to visualize the underlying exponential law within the original time series (Wei, Xie, & Yang, 2020; Xie, 2022); (2) grey models possess the capability to extrapolate their dynamic trends by analyzing increments and growth rates of sequences (Li & Xie, 2023); and (3) grey models usually exhibit straightforward structures and fewer parameters, eliminating the need for extensive sample data to train repeatedly (Zhou & Ding, 2021). For these reasons, grey models have been successfully applied in various fields (Liu, Tao, Xie, Tao, & Hu, 2022).

The best-known grey model, GM(1,1), utilizes a first-order (the first ‘1’) differential equation with a single variable (the second ‘1’) to forecast quasi-exponential time series. However, it is hard to capture the seasonal patterns present in the data. To overcome this issue, various seasonal grey models have been developed. For example, the discrete grey seasonal models incorporate a cycle truncation accumulated generating operator to the discrete grey model for forecasting fashion retail demand (Xia & Wong, 2014) and traffic flow (Xiao, Yang, Mao, & Wen, 2017). Seasonal factors are integrated into the accumulative sum operator of grey models to reflect seasonal effects (Su, Yan, Wu, & Zeng, 2022; Wang et al., 2018). Data grouping approaches are also adopted in grey modeling (Wang, Li, & Pei, 2017). While these models use some de-seasonalization approaches to address seasonality, they still fail to identify essential seasonal characteristics. Some recent grey models incorporate dummy variables to represent seasonal fluctuations (Zhou, Cheng, Ding, Chen, & Li, 2021; Zhou, Jiang et al., 2021), but this approach requires numerous seasonal variables and may be prone to problems related to degrees of freedom. Additionally, several grey models introduce trigonometric functions into their forcing term, such as GM(1, 1|sin) (Mao, Chen, & Xiao, 2012) and GM(1, 1|cos, sin) (Comert et al., 2021). However, these models are not adaptive for complex multi-frequency seasonal patterns. Recent work by Wang et al. (2022) employed Fourier series to approximate seasonal forcing, but did not account for the time-varying growth rates since it used a fixed development coefficient.

In this paper, we extend the existing grey model framework by introducing Fourier time-varying parameters that more flexibly characterize the seasonal variability in time series forecasting.

3. Fourier time-varying grey model

Here, we present the mathematical formulation of our grey forecasting model. Our model is an integro-differential equation with time-varying parameters, derived through the continuous-time grey model using integral matching. Within this model, we first address the seasonal variations observed in product demand data and use a Fourier series to characterize its pattern. Then, we introduce an integral transformation operator and a least squares criterion to estimate the model parameters. Finally, we outline a strategy for selecting the appropriate Fourier order.

3.1. Construction of Fourier time-varying grey model

Assuming that $X(t) = \{x(t_1), x(t_2), \ldots, x(t_m)\}$ is the original time series, the accumulative sum series is defined as

$$Y(t) = \{y(t_1), y(t_2), \ldots, y(t_m)\}, \tag{1}$$

where

$$y(t_k) = \sum_{i=1}^{k} h_i x(t_i), \quad h_k = \begin{cases} 1, & k = 1 \\ t_k - t_{k-1}, & k \geq 2 \end{cases} \tag{2}$$

GM(1,1) with time-varying parameters for the accumulating generation series is defined by the following differential equation:

$$\begin{cases} \frac{d}{dt}y(t) = a(t)y(t) + b(t), & t \geq t_1, \\ y(t_1) = \eta_y. \end{cases} \tag{3}$$

where $a(t)$, referred to as the development coefficient, depicts the dynamics of growth rates in the demand series over time; $b(t)$ is the forcing term whose value varies over time; and η_y is an unknown initial value of the accumulative sum series $Y(t)$.

Eq. (3) models the accumulative sum series and leverages the inverse accumulative generating operator to obtain the fitting and forecasting values of the original series. However, this process inevitably leads to additional modeling errors. A recent study by Wei et al. (2020) demonstrates that integral matching can realize direct modeling from the original time series. Following this approach, we propose an alternative model equation.

By integration

$$y(t) = \eta_y + \int_{t_1}^{t} x(s)ds,\ \ t \geq t_1, \tag{4}$$

Eq. (3) is equivalent to the reduced-order integro-differential equation

$$\begin{cases} \frac{d}{dt}x(t) = a(t)x(t) + a'(t)\left(\eta_y + \int_{t_1}^{t} x(s)ds\right) + b'(t),\ t \geq t_1, \\ x(t_1) = \eta_x. \end{cases} \tag{5}$$

where $\eta_x = a(t_1)\eta_y + b(t_1)$ is the unknown initial value of original time series, and $a'(t)$ and $b'(t)$ are the derivatives of the time-varying parameters $a(t)$ and $b(t)$, respectively.

It is crucial to select an appropriate function to model the variation of parameters, since the performance of the time-varying parametric model depends heavily on the choice of this varying function. In general, Fourier functions, polynomial functions, and basis functions can be used to fit time-varying parametric series models. A Fourier series, which comprises a summation of cosine and sine functions with increasing frequencies, is effective for revealing the periodicity of coefficients over time (Vu, Muttaqi, Agalgaonkar, & Bouzerdoum, 2017). A finite Fourier series can be used to approximate seasonal patterns (Becker, Enders, & Hurn, 2006). In our study, we opt for a finite Fourier series to approximately characterize the time-varying parameters. The typical form of the time-varying parameters applied to the finite Fourier series is given as

$$a(t) = \alpha_0 + \sum_{n=1}^{N}\Big(\alpha_{1n}\cos(n\omega t) + \alpha_{2n}\sin(n\omega t)\Big), \tag{6}$$

$$b(t) = \beta_0 + \sum_{n=1}^{N}\Big(\beta_{1n}\cos(n\omega t) + \beta_{2n}\sin(n\omega t)\Big). \tag{7}$$

where N is the number of frequencies contained in the approximation, and when N tends to infinity, $a(t)$ and $b(t)$ can represent any periodic functions; $\omega = \frac{2\pi}{T}$ is the angular frequency and T is the seasonal period; $\alpha_0 = \frac{1}{T}\int_0^T a(t)dt$ and $\beta_0 = \frac{1}{T}\int_0^T b(t)dt$ are the first coefficients of $a(t)$ and $b(t)$, respectively; and $\alpha_{1n} = \frac{2}{T}\int_0^T a(t)\cos(n\omega t)dt$, $\alpha_{2n} = \frac{2}{T}\int_0^T a(t)\sin(n\omega t)dt$, $\beta_{1n} = \frac{2}{T}\int_0^T b(t)\cos(n\omega t)dt$, and $\beta_{2n} = \frac{2}{T}\int_0^T b(t)\sin(n\omega t)dt$ are the Fourier coefficients.

Substituting Eqs. (6) and (7) into Eq. (5) yields the Fourier time-varying grey model, termed FTGM(1,1,N). Then, the time-varying parameters can be gathered in the parameter set, which can be expressed in the form of a vector: $\boldsymbol{\theta} = [\alpha_0\ \beta_0\ \alpha_{11}\ \alpha_{21} \cdots \alpha_{1N}\ \alpha_{2N}\ \beta_{11}\ \beta_{21} \cdots \beta_{1N}\ \beta_{2N}]^{\top}$.

3.2. Integral matching for parameter estimation

In this section, we introduce integral matching, which consists of an integral transformation operator and a least squares criterion, to estimate the model's structural parameters.

By integration, Eq. (5) yields the integral equation (derivation see Appendix A)

$$\begin{cases} x(t) = a(t)\left(x(t_1) + \int_{t_1}^{t} x(s)ds\right) + b(t), \quad t \geq t_1, \\ x(t_1) = \frac{b(t_1)}{1-a(t_1)}. \end{cases} \tag{8}$$

where $a(t) = \alpha_0 + \sum_{n=1}^{N}(\alpha_{1n}\cos(n\omega t) + \alpha_{2n}\sin(n\omega t))$ and $b(t) = \beta_0 + \sum_{n=1}^{N}(\beta_{1n}\cos(n\omega t) + \beta_{2n}\sin(n\omega t))$.

Using the piecewise linear quadrature, the definite integral terms of Eq. (8) can be approximated by

$$\begin{aligned} \int_{t_1}^{t_k} x(s)ds &= \frac{1}{2}\sum_{i=2}^{k} h_i x(t_{i-1}) + \frac{1}{2}\sum_{i=2}^{k} h_i x(t_i) \\ &= \frac{1}{2}\sum_{i=2}^{k} h_i x(t_{i-1}) + \frac{1}{2}y(t_k) - \frac{1}{2}x(t_1) \end{aligned} \tag{9}$$

In particular, for equally spaced time series, Eq. (9) yields

$$\int_{t_1}^{t_k} x(s)ds = \frac{1}{2}y(t_{k-1}) + \frac{1}{2}y(t_k) + \frac{h-2}{2}x(t_1) \tag{10}$$

Substituting Eq. (10) into Eq. (8) gives the corresponding discrete-time equation:

$$x(t_k) = a(t_k)\left(\frac{1}{2}y(t_{k-1}) + \frac{1}{2}y(t_k) + \frac{h}{2}x(t_1)\right) + b(t_k) + \varepsilon(t_k) \tag{11}$$

where $\varepsilon(t_k)$ is the sum of discretization and measurement errors with mean 0 and variance σ_ε^2. Then, the reduced discrete-time equation (11) is equivalent to

$$x(t_k) = a(t_k)z(t_k) + b(t_k) + \varepsilon(t_k) \tag{12}$$

where $z(t_k) = \frac{1}{2}y(t_{k-1}) + \frac{1}{2}y(t_k) + \frac{h}{2}x(t_1)$, which is a linear combination of the adjacent accumulative sum series $y(t_{k-1})$ and $y(t_k)$, as well as the initial value $x(t_1)$ of the original series.

Based on the Fourier series Eqs. (6) and (7) for time-varying parameters $a(t_k)$ and $b(t_k)$, the discrete-time equation (12) can be further formulated as

$$\begin{aligned} x(t_k) &= \alpha_0 z(t_k) + \beta_0 \\ &\quad + \sum_{n=1}^{N}\Big(\alpha_{1n}\cos(n\omega t_k) + \alpha_{2n}\sin(n\omega t_k)\Big) z(t_k) \\ &\quad + \sum_{n=1}^{N}\Big(\beta_{1n}\cos(n\omega t_k) + \beta_{2n}\sin(n\omega t_k)\Big) + \varepsilon(t_k) \end{aligned} \tag{13}$$

or in vector form:

$$\boldsymbol{X} = \Xi(\boldsymbol{z}, \boldsymbol{f})\boldsymbol{\theta} + \boldsymbol{\varepsilon} \tag{14}$$

where

$$\mathbf{X} = \begin{bmatrix} x(t_2) \\ x(t_3) \\ \vdots \\ x(t_m) \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon(t_2) \\ \varepsilon(t_3) \\ \vdots \\ \varepsilon(t_m) \end{bmatrix},$$

$$\Xi(\boldsymbol{z}, \boldsymbol{f}) = \begin{bmatrix} z(t_2) & 1 & \boldsymbol{f}(t_2)z(t_2) & \boldsymbol{f}(t_2) \\ z(t_3) & 1 & \boldsymbol{f}(t_3)z(t_3) & \boldsymbol{f}(t_3) \\ \vdots & \vdots & \vdots & \vdots \\ z(t_m) & 1 & \boldsymbol{f}(t_m)z(t_m) & \boldsymbol{f}(t_m) \end{bmatrix},$$

$$\boldsymbol{f} = \begin{bmatrix} \cos(\omega t_2) & \sin(\omega t_2) & \cdots & \cos(nwt_2) & \sin(nwt_2) \\ \cos(\omega t_3) & \sin(\omega t_3) & \cdots & \cos(nwt_3) & \sin(nwt_3) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \cos(\omega t_m) & \sin(\omega t_m) & \cdots & \cos(nwt_m) & \sin(nwt_m) \end{bmatrix},$$

$$\boldsymbol{z} = \begin{bmatrix} \frac{1}{2}y(t_1) + \frac{1}{2}y(t_2) + \frac{h}{2}x(t_1) \\ \frac{1}{2}y(t_2) + \frac{1}{2}y(t_3) + \frac{h}{2}x(t_1) \\ \vdots \\ \frac{1}{2}y(t_{m-1}) + \frac{1}{2}y(t_m) + \frac{h}{2}x(t_1) \end{bmatrix}.$$

Minimizing the criterion function $\Theta(\boldsymbol{\theta}) = \|\boldsymbol{X} - \Xi(\boldsymbol{z}, \boldsymbol{f})\boldsymbol{\theta}\|_2^2$ with respect to the parameter $\boldsymbol{\theta}$ yields the estimators

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \left[\hat{\alpha}_0, \hat{\beta}_0, \hat{\alpha}_{11}, \hat{\alpha}_{12}, \ldots, \hat{\alpha}_{n1}, \hat{\alpha}_{n2}, \hat{\beta}_{11}, \hat{\beta}_{12}, \ldots, \hat{\beta}_{n1}, \hat{\beta}_{n2}\right]^\top \\ &= \left(\Xi(\boldsymbol{z}, \boldsymbol{f})^\top \Xi(\boldsymbol{z}, \boldsymbol{f})\right)^{-1} \Xi(\boldsymbol{z}, \boldsymbol{f})^\top \mathbf{X} \end{aligned} \tag{15}$$

Note that this expression for the parameter vector is well defined only if the inverse matrix in Eq. (15) exists.

Unfortunately, no analytical solution exists for Eq. (5), meaning that the integro-differential equation has to be solved numerically. In this context, we turn to the widely used `ODE45` solver in `MATLAB`,[1] which serves as the standard solver for first-order differential equations. The `ODE45` solver implements the fourth-order Runge–Kutta algorithm with a fifth-order truncation error and is a preferred method for numerical solutions due to its high accuracy (Ahmadianfar, Heidari, Gandomi, Chu, & Chen, 2021; Dormand & Prince, 1980).

[1] Alternatively, one can also solve the numerical solution of Eq. (5) by using the `odeint` function from the `SciPy` library in `Python` or the `ode()` function of the `deSolve` package in `R`.

3.3. Order selection of Fourier time-varying grey model

Upon determining an appropriate time-varying function to represent seasonal variations, as described in Section 3.1, the choice of Fourier order becomes critical for capturing seasonal patterns. Opting for a small order value might be inadequate to depict the full range of seasonal variation characteristics, while a high order value can increase model complexity and introduce the risk of overfitting. Therefore, a logical approach to determining a suitable Fourier order is to assess model performance using various candidate orders. Guiding this selection is the Nyquist–Shannon sampling theorem (Por, van Kooten, & Sarkovic, 2019), which offers valuable insights into identifying these candidate orders. Specifically, given a sampling interval of h_k and a seasonal period of T, the sampling frequency is $1/h_k$. According to the Nyquist–Shannon sampling theorem, the signal's highest frequency in the Fourier series is N/T, and the alternative order is suggested to be $N < T/(2h_k)$. This indicates that a continuous signal can be perfectly recovered when sampled at a rate exceeding the signal's highest frequency by at least twice.

Algorithm 1: Fourier order selection algorithm for FTGM(1,1,N).

Input : Seasonal time series $X = \left\{x(t_1), x(t_2), \cdots, x(t_g)\right\}$, seasonal period T

Output: Best Fourier order N_{opt} and forecasted data $\hat{X}_{\text{test}}$

Divide X into X_{train}, X_{valid} and X_{test}:
  $X_{\text{train}} = \{x(t_1), x(t_2), \cdots, x(t_m)\}$;
  $X_{\text{valid}} = \{x(t_{m+1}), x(t_{m+2}), \cdots, x(t_{m+l})\}$;
  $X_{\text{test}} = \left\{x(t_{m+l+1}), x(t_{m+l+2}), \cdots, x(t_{m+l+J})\right\}$
Determine the candidate order
  $N \in \mathcal{P} = \{1, 2, \cdots, \lceil T/(2h_k)\rceil - 1\}$, where h_k is the sampled interval, and $\lceil x \rceil$ is the least integer that is greater than or equal to x
for $N \in \mathcal{P}$ **do**
  $\boldsymbol{\theta}_N \leftarrow \underset{\boldsymbol{\theta}_N, \boldsymbol{\theta}_N \neq 0}{\arg\min} \left\{\Theta(\boldsymbol{\theta}_N) = \|\boldsymbol{X}_{\text{train}} - \Xi\boldsymbol{\theta}_N\|^2\right\}$
  $\hat{X}_{\text{train}} = \left\{\hat{x}(t_1), \hat{x}(t_2), \cdots, \hat{x}(t_m)\right\} \leftarrow$ Fit the X_{train} using $\boldsymbol{\theta}_N$
  $\hat{X}_{\text{valid}} = \left\{\hat{x}(t_{m+1}), \hat{x}(t_{m+2}), \cdots, \hat{x}(t_{m+l})\right\} \leftarrow$ Forecast the X_{valid} using $\boldsymbol{\theta}_N$
  $\text{RMSE}_{\text{train}}(N) \leftarrow$ Evaluate the fitted errors using RMSE
  $\text{RMSE}_{\text{valid}}(N) \leftarrow$ Evaluate the validation errors using RMSE
end
$N_{\text{opt}} = \underset{N \in \mathcal{P}}{\arg\min} \{\text{RMSE}_{\text{valid}}(N)\}$
$\hat{X}_{\text{test}} \leftarrow$ Forecast the X_{test} using FTGM(1,1,N_{opt});
return $N_{\text{opt}}, \hat{X}_{\text{test}}$.

To identify the optimal Fourier order among various alternatives, we propose a selection strategy based on a validation set approach. The dataset is divided into three parts: a training set, a validation set, and a test set. The

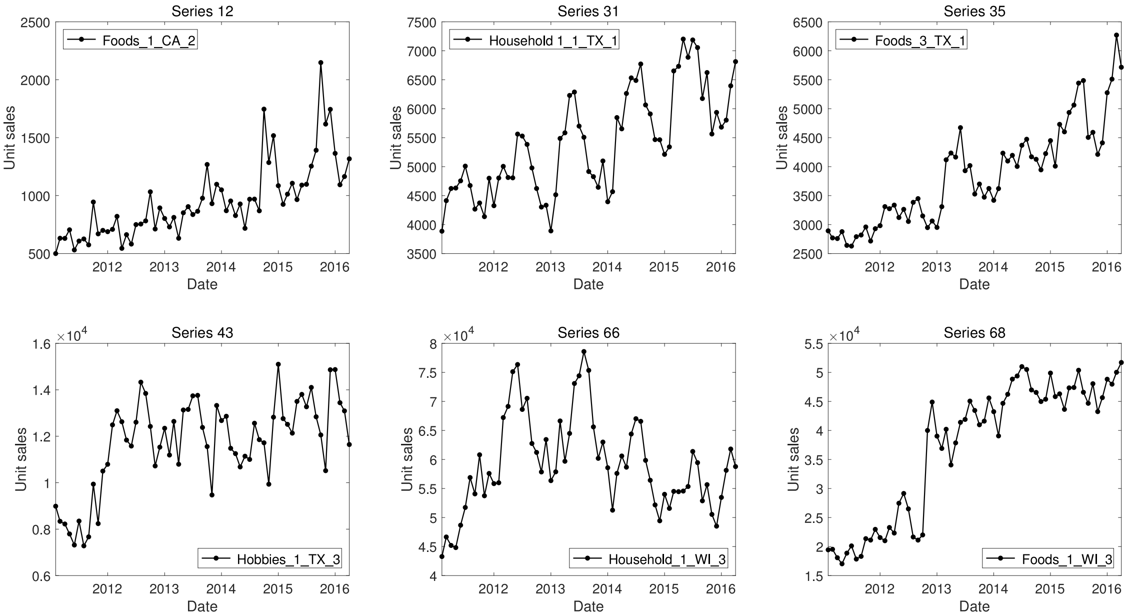

Fig. 1. A sample of series reporting monthly unit sales at the department-store level. 'Foods_1_CA_2' refers to the sales recorded for the product of the *Foods 1* department sold at the second store in *CA*, and so on.

training set is used to fit the models and estimate parameters, the validation set is used to select the model order, and a test set is used for evaluating model forecasting performance. In this study, we employ the root mean squared error (RMSE) as the preferred error measure for choosing the model order, due to its consistency with the loss function (mean squared error) used for model parameter estimation. The best Fourier order selection step is shown in Algorithm 1.

4. Experimental design

In this section, we describe the dataset, formulate benchmark models, define three error metrics, and present our rolling origin evaluation approach. We also elucidate model training processes and estimated parameters.

4.1. Data

To evaluate the forecasting performance of our proposed model on real-world data, we conduct experiments using a dataset from the M5 competition (Makridakis, Spiliotis, & Assimakopoulos, 2022b). We use this dataset because our initial motivation comes from a retail context, and the M5 contains sales data from Walmart. This dataset features a natural hierarchy, covering the sales of 10 stores located in three USA states, across seven product departments (Hobbies_1, Hobbies_2, Household_1, Household_2, Foods_1, Foods_2, and Foods_3). To demonstrate that our results are not biased in terms of the product departments or stores, we consider all 70 time series at the department-store level. To both avoid intermittence and visualize seasonal patterns in small sample data, we aggregate the daily series (selling history of 1941 days) to a monthly frequency. This yields an aggregated dataset comprising 70 distinct products, each with 63 observations. The corresponding series names are presented in Appendix B.

Fig. 1 illustrates six representative series of monthly unit sales at the department-store level, showcasing typical characteristics present in the dataset. As seen, each series exhibits distinctive time-varying seasonality. Some series display strong seasonality (e.g., series 12), while others exhibit partially stable (e.g., series 31) or unstable seasonal patterns (e.g., series 43). Seasonal trends also vary across series: some show rising trends with different degrees of amplitude over time, while others depict an initial increase followed by a subsequent decrease in seasonal trends (e.g., series 66). Moreover, certain series feature noticeably abrupt changes that break the intrinsic seasonal patterns (e.g., series 35 and 68). Consequently, effectively forecasting these time series poses a significant challenge.

4.2. Benchmark models

To compare our model performance with other state-of-the-art methods, we consider 11 benchmark models in our study. Among seasonal grey models, three common strategies are typically utilized to deal with seasonality, namely dummy variables, data grouping, and seasonal factors. Accordingly, we select seasonal grey models that employ these three techniques as our benchmarks. Additionally, seasonal time series forecasting models, including traditional statistical methods and more recently developed statistical ones (such as Fourier representations of seasonality), are chosen as benchmarks. Two machine learning methods often applied in time series forecasting are also included for comparison with the proposed FTGM.

(1) **Seasonal grey model (SGM)**: SGM(1,1), proposed by Wang et al. (2018), uses a seasonal factor to generate the accumulation operation, expressed as $y_s(k) = \sum_{i=1}^{k} x(i)/f_s(i)$. The seasonal factor $f_s(i)$ quantifies the average deviation of the actual value from the trend value in the current period, attributable to seasonal effects. The expression for the seasonal factor is given by

$$f_s(i) = \frac{n\sum_{(k-1) \bmod s+1=i}^{k=1,2,\ldots,n} x(k)}{\left\lfloor \frac{n-i+s}{s} \right\rfloor \sum_{j=1}^{n} x(j)},\quad i = 1,2,\ldots,s \tag{16}$$

where s represents the seasonal period, and $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z. The grey differential equation for SGM(1,1) is

$$\frac{dy_s(t)}{dt} = a_s y_s(t) + b_s \tag{17}$$

and the parameters a_s and b_s are estimated using ordinary least squares.

(2) **Data grouping grey model (DGGM)**: The DGGM collects data from the same seasons into a group set, and then employs GM(1,1) to model the time series in each grouped set (Wang et al., 2017). The time series data are divided according to cycles: $x_i = (x_i(1), x_i(2), \ldots, x_i(n))$, $i = 1, 2, \ldots, s$ (where s denotes the seasonal period).

(3) **Discrete grey seasonal model (DGSM)**: The DGSM enables the incorporation of seasonality by introducing seasonal dummy variables, as outlined in the work by Zhou and Ding (2021). Its expression is denoted as

$$y(k+1) = \alpha y(k) + \beta_{M(k+1,s)} \tag{18}$$

where s is the seasonal period, $\beta_{M(k+1,s)}$ represents the seasonal component, and

$$M(k,s) = \begin{cases} s, & k \bmod s = 0, \\ k \bmod s, & k \bmod s \neq 0, \end{cases} \tag{19}$$

(4) **Seasonal naive model (SNaive)**: Similar to the naive method, SNaive forecasts the value at time $t+h$ equal to the last known observation from the same season of the year. Unlike the naive method, SNaive can capture possible seasonal variations in the data (Hyndman & Athanasopoulos, 2021). The expression for SNaive is given by

$$\hat{x}_{t+h|t} = x_{t+h-m(k+1)} \tag{20}$$

where m is the seasonal period, and k is the integer part of $(h-1)/m$, indicating the number of complete years in the forecast period prior to time $t+h$.

(5) **Seasonal autoregressive integrated moving average (SARIMA)**: The SARIMA method is an extension of the popular ARIMA model that includes three additional hyperparameters for a fixed seasonal cycle, which facilitates the modeling of a large variety of seasonal data (Wolters & Huchzermeier, 2021).

(6) **Holt–Winters exponential smoothing (HWES)**: The HWES method, proposed by Holt (2004) and Winters (1960), has been enhanced to account for seasonality within the data. The method comprises a forecasting equation and three smoothing equations (the level ℓ_t, the trend b_t, and the seasonal component s_t), with corresponding smoothing parameters α, β, and γ. The equations are

$$\begin{aligned} \hat{x}_{t+h|t} &= \ell_t + hb_t + s_{t+h-m(k+1)} \\ \ell_t &= \alpha(x_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\ b_t &= \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1} \\ s_t &= \gamma(x_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m} \end{aligned} \tag{21}$$

where m denotes the frequency of the seasonality, and k is the integer part of $(h-1)/m$.

(7) **Dynamic harmonic regression autoregressive integrated moving average (DHR-ARIMA)**: The DHR-ARIMA model utilizes Fourier terms to portray seasonal effects and uses ARIMA errors to model transient time series dynamics. As noted by Young, Pedregal, and Tych (1999), the DHR model represents a decomposition that contains the trend, seasonal, cyclical, and white noise components:

$$x_t = T_t + S_t + C_t + e_t \tag{22}$$

Here, the most important unobserved components are the seasonal term S_t and the cyclical term C_t, which are defined by

$$\text{Seasonal}: S_t = \sum_{i=1}^{R_s} \left\{ a_{i,t}\cos(\omega_i t) + b_{i,t}\sin(\omega_i t) \right\} \tag{23}$$

and

$$\text{Cyclical}: C_t = \sum_{i=1}^{R_c} \left\{ \alpha_{i,t}\cos(f_i t) + \beta_{i,t}\sin(f_i t) \right\} \tag{24}$$

where $a_{i,t}$, $b_{i,t}$, $\alpha_{i,t}$, and $\beta_{i,t}$ are time-varying parameters, and ω_i and f_i are frequencies associated with the seasonality and cyclical component in the series, respectively.

(8) **Exponential smoothing state-space model with Box–Cox transformation, ARMA errors, and trend and seasonal components (BATS)**: The BATS model extends the traditional seasonal innovations models to allow for multiple seasonal periods (De Livera et al., 2011). The Box–Cox transformation serves for dealing with non-linear data, while the ARMA modeling component is employed to address autocorrelated residuals in time series data. The notation x_t, with parameter ω, denotes the Box–Cox transformed observations, and is expressed as

$$x_t^{(\omega)} = \begin{cases} \frac{x_t^{\omega}-1}{\omega}, & \omega \neq 0, \\ \log x_t, & \omega = 0, \end{cases}$$

$$x_t^{(\omega)} = \ell_{t-1} + \phi b_{t-1} + \sum_{i=1}^{T} s_{t-m_i}^{(i)} + d_t,$$

$$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha d_t,$$
$$b_t = (1-\phi)b + \phi b_{t-1} + \beta d_t,$$
$$s_t^{(i)} = s_{t-m_i}^{(i)} + \gamma_i d_t, \tag{25}$$
$$d_t = \sum_{i=1}^{p} \varphi_i d_{t-i} + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \varepsilon_t,$$

where $m_1, \ldots, m_T$ represent the seasonal periods, ℓ_t denotes the local level at time t, b_t corresponds to the short-term trend at time t, b corresponds to the long-term trend, $s_t^{(i)}$ represents the ith seasonal component at time t, d_t denotes an ARMA(p,q) process, and ε_t stands for a Gaussian white noise process with a zero mean and constant variance σ^2.

(9) **Trigonometric Box–Cox ARMA trend seasonal (TBATS) model**: The TBATS model extends the BATS model by incorporating trigonometric seasonal terms (De Livera et al., 2011). This trigonometric representation of seasonality effectively reduces the number of model parameters in cases with high-frequency seasonal patterns. Additionally, this approach enhances the model's flexibility to handle complex seasonal patterns. The trigonometric Fourier model can be expressed as

$$s_t^{(i)} = \sum_{j=1}^{k_i} s_{j,t}^{(i)}$$
$$s_{j,t}^{(i)} = s_{j,t-1}^{(i)} \cos \lambda_j^{(i)} + s_{j,t-1}^{*(i)} \sin \lambda_j^{(i)} + \gamma_1^{(i)} d_t \tag{26}$$
$$s_{j,t}^{*(i)} = -s_{j,t-1} \sin \lambda_j^{(i)} + s_{j,t-1}^{*(i)} \cos \lambda_j^{(i)} + \gamma_2^{(i)} d_t$$

where $\gamma_1^{(i)}$ and $\gamma_2^{(i)}$ correspond to smoothing parameters, and $\lambda_j^{(i)} = 2\pi j/m_i$. $s_{j,t}^{(i)}$ denotes the stochastic *level* of the ith seasonal component, $s_{j,t}^{*(i)}$ represents the stochastic *growth* in the *level* of the ith seasonal component, and k_i indicates the number of harmonics required for the ith seasonal component.

(10) **Multi-layer perceptron (MLP)**: The MLP is a feed-forward neural network widely used for tasks such as function approximation, pattern classification, process control, and time series forecasting (Lima-Junior & Carpinetti, 2019). Comprising multiple layers of neurons, an MLP includes one input layer, one or more intermediate hidden layers, and an output layer (Cruz, Costa, Krippahl, & Lopes, 2022). Training the MLP employs the backpropagation algorithm, encompassing forward propagation (computing predictions) and backpropagation (updating weights and biases). The one-step-ahead forecast $\hat{y}_{t+1}$ is formulated using lagged observations of the time series or other explanatory variables as inputs, and is defined by Kourentzes, Barrow, and Crone (2014):

$$\hat{y}_{t+1} = \beta_0 + \sum_{h=1}^{H} \beta_h g\left(\gamma_{0i} + \sum_{i=1}^{I} \gamma_{hi} p_i\right). \tag{27}$$

where I is the number of inputs p_i of the *NN*, and H is the number of hidden nodes. $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_H]$ are weights from the hidden to the output layers, and $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_{HI}]$ are weights from the input to the hidden layers. β_0 and γ_{0i} are the biases of each neuron, which for each neuron act similarly to the intercept in a regression. $g(\cdot)$ is a non-linear transfer function, which is usually either the sigmoid logistic or the hyperbolic tangent function.

(11) **Long short-term memory (LSTM)**: LSTM, as a particular type of RNN, is capable of capturing long-term dependencies in sequences (Twumasi & Twumasi, 2022). In contrast to the conventional perceptron architecture, LSTM incorporates a cell to retain temporal network states and utilizes gating mechanisms for information regulation. These gates include an input gate, a forget gate, and an output gate (Greff, Srivastava, Koutnik, Steunebrink, & Schmidhuber, 2017). Specifically, the input gate determines new information to be stored in the cell state, the forget gate eliminates irrelevant information in the cell state, and the output gate decides on the produced output. The operation of LSTM is as follows:
First, calculate the input gate (a_t), forget gate (f_t), and output gate (o_t) based on the current input (x_t) and previous hidden state (h_{t-1}):

$$f_t = \sigma(w_f x_t + w_{hf} h_{t-1} + b_f)$$
$$a_t = \sigma(w_a x_t + w_{ha} h_{t-1} + b_a) \tag{28}$$
$$o_t = \sigma(w_o x_t + w_{ho} h_{t-1} + b_o)$$

Second, calculate the candidate value (z_t) using a tanh activation layer:

$$z_t = \tanh(w_c x_t + w_{hc} h_{t-1} + b_c) \tag{29}$$

Then, update the state value (c_t) by combining the results of the forget gate (f_t) and input gate (a_t):

$$c_t = f_t \times c_{t-1} + a_t \times z_t \tag{30}$$

Finally, calculate the new hidden state (h_t) by combining the cell state (c_t) and the filtered output (o_t):

$$h_t = o_t \times \tanh(c_t) \tag{31}$$

In these equations, σ and *tanh* are activation functions, and $\times$ indicates point-wise multiplication. $b = \{b_a, b_f, b_c, b_o\}$ are biases of the input gate, forget gate, internal state, and output gate; $\vec{w}_1 = \{w_a, w_f, w_c, w_o\}$ are weight matrices of the input gate, forget gate, internal state, and output gate; $\vec{w}_2 = \{w_{ha}, w_{hf}, w_{hc}, w_{ho}\}$ are the recurrent weights; $\vec{a}_1 = \{a(t_i), f(t_i), c(t_i), o(t_i)\}$ are the output results for the input gate, forget gate, internal state, and output gate.

4.3. Error metrics

In this section, we evaluate the forecasting performance of both the proposed model and the benchmark models using different error metrics. In addition to one traditional error measure, the root mean squared error (RMSE), we also adopt two scale-independent metrics used in the M5 competition (Makridakis et al., 2022b), namely the mean absolute scaled error (MASE) and the root mean squared scaled error (RMSSE). We do not use the weighted RMSSE (WRMSSE) employed in the M5 competition, as the price weighting of the RMSSE tends to reduce its stability at higher aggregation levels, including the department-store level (Hewamalage, Montero-Manso, Bergmeir, & Hyndman, 2021).

The RMSE is a commonly used scale-dependent measure, due to its straightforward calculation and clear interpretation. It also holds practical meaning for retailing managers, as they prioritize fast-moving products that contribute more revenue than slow-moving items in stores (Ma, Fildes, & Huang, 2016). It is given by

$$\mathrm{RMSE} = \sqrt{\frac{1}{J}\sum_{j=1}^{J}\Big(x(t_j) - \hat{x}(t_j)\Big)^2} \tag{32}$$

where J represents the forecast horizon, $x(t)$ denotes the actual value of the series at time t, and $\hat{x}(t)$ represents the forecast for time t.

The MASE, proposed by Hyndman and Koehler (2006), can be considered as a weighted arithmetic mean of the MAE based on the variations of the sales data in the estimation period (Davydenko & Fildes, 2013). It is defined as

$$\mathrm{MASE} = \frac{1}{J}\frac{\sum_{j=1}^{J}\left|x(t_j) - \hat{x}(t_j)\right|}{\frac{1}{n-1}\sum_{k=2}^{n}|x(t_k) - x(t_{k-1})|} \tag{33}$$

where n is the number of in-sample observations.

The RMSSE, an MASE variant, is advocated for assessing forecasting accuracy in the M5 competition (Makridakis, Spiliotis, & Assimakopoulos, 2022a). The RMSSE is defined as follows:

$$\mathrm{RMSSE} = \sqrt{\frac{\frac{1}{J}\sum_{j=1}^{J}\left(x(t_j) - \hat{x}(t_j)\right)^2}{\frac{1}{n-1}\sum_{k=2}^{n}(x(t_k) - x(t_{k-1}))^2}} \tag{34}$$

where the notation is unchanged.

To measure the performance of the competing models across the series, we further calculate the mean and median values of these primary error measures across the series. Because the RMSE is a scale-dependent error measure, averaging the RMSE is influenced by the scale of different sequences. Thus, we only compute the mean and median of MASEs and the mean and median of RMSSEs.

4.4. Rolling origin evaluation

In this study, we employ the rolling origin evaluation for assessing the overall performance of the proposed FTGM and benchmark models. This technique contributes to generating robust forecasts by averaging performance scores from each origin, thereby reducing sensitivity to random occurrences (Svetunkov, 2023).

All time series from the monthly M5 dataset have 63 observations, which are split into two periods: 48 months for training, and 15 months for forecasting comparison. To implement this rolling origin evaluation, given a rolling origin of six, we begin with an initial in-sample set comprising 43 observations for each time series. The last observation in the in-sample set is referred to as the initial forecast origin. We utilize this initial in-sample set to estimate the parameters of each forecasting model and produce forecasts for the subsequent 15 periods, thereby setting the forecast horizon to 15 steps ahead. The performance evaluation involves comparing the model's predicted values for periods 44 to 58 with the corresponding actual observations, using the metrics specified in Section 4.3 of this study. Subsequently, we expand the in-sample set by one observation, resetting the forecast origin to period 44. We then re-estimate the model parameters and generate forecasts for the subsequent horizon, covering periods 45 to 59, followed by performance measurements. This iterative process continues by incrementally adding one period at a time until we reach the end of the available data. The final forecast origin is established at period 48, and forecasts for periods 49 to 63 are generated accordingly. Fig. 2 illustrates such a setup. In total, we produce six one- to 15-step-ahead forecasts, ranging from origins 43 to 48, for each of the 70 time series. The performance scores computed in each horizon of the six origins are averaged to estimate the overall performance of the forecasting models.

4.5. Model training

For the selection of the Fourier order in the FTGM, we use the method presented in Algorithm 1. Given the monthly period $T = 12$ and sampled interval $h_k = 1$, the alternative orders are recommended as N=1, 2, 3, 4, 5 (where $N \in \mathcal{P} = \{1, 2, \ldots, \lceil T/(2h_k)\rceil - 1\}$). We construct the FTGM for each of the 70 monthly M5 series using the recommended alternative orders. The data series is split into three parts: 42 months for training, the middle six months for validation, and the remaining 15 months for forecasting. We individually select an appropriate Fourier order for each origin based on $\mathrm{RMSE}_{\mathrm{valid}}$. After the optimal Fourier order is determined, we estimate the model parameters by integral matching and the least squares methods, as detailed in Section 3.2. For different forecast origins, we repeat the process of selecting the optimal Fourier order and estimating model parameters. The FTGM is implemented using`Matlab` programming language.

For the grey models being compared, the parameters are calculated based on the least squares method, which is implemented on the `Matlab` platform. The SNaive, HWES, SARIMA, BATS, and TBATS models are implemented using the `SNaive()`, `ets()`, `auto.arima()`, `bats()`, and `tbats()` functions, respectively, from the `forecast` package in `R` (Hyndman & Khandakar, 2008). The DHR-ARIMA model is implemented by combining the `auto.arima()` and `fourier()` functions. Here, the parameter estimates and model (order) selection are determined based on likelihood functions and the Akaike information criterion. The MLP is implemented by using `mlp()` functions in the `nnfor` package of `R` (Kourentzes, 2022). In the MLP, five-fold cross-validation is used to automatically determine both the number of hidden nodes and the differencing order of the training samples. For LSTM, we utilize Python's `Keras` to build the model, comprising one input layer,

Fig. 2. Rolling origin applied for evaluating the model performance.

one LSTM layer with memory blocks, and one output layer with a dimensionality of 15. The input layer employs the `tanh` function as its activation function with a time step of 12. The number of hidden units is set to 24 by trial and error. We employ the `adam` optimizer and `RMSE` as the loss function for model compilation. During model training, we use the minibatch gradient descent with a fixed learning rate of 0.001, operating on batches of six sequences over 20 epochs. To mitigate overfitting, we incorporate a 50% dropout.

Table C.1 of Appendix C summarizes the estimated parameters for the FTGM and benchmarks. The data are from Hobbies_1_CA_1 product monthly sales and the first forecast origins.

5. Results and discussion

This section details the empirical results in terms of accuracy, ranks, statistical significance, and an ablation study, and then gives the evaluation results on relatively shorter time series.

5.1. Accuracy comparisons

Table 1 presents the forecasting results of the proposed model and benchmarks for h-step-ahead forecast horizons. For each horizon, the results of the best-performing methods are highlighted in bold. Across all forecast horizons, the proposed FTGM consistently outperforms the benchmark models in terms of both mean/median MASE and mean/median RMSSE metrics. For long forecast horizons ($h = 9$, 12, and 15), SGM and DGSM perform better than the other benchmark models. SGM is better than the rest of the benchmarks in mean MASE and mean RMSSE for $h = 6$, while DGSM demonstrates better performance in terms of the median MASE and median RMSSE. For the short forecast horizon ($h = 1$), BATS outperforms the others in terms of the mean MASE, median MASE, and median RMSSE metrics, and for $h = 3$, DGSM outperforms all other benchmarks.

Fig. 3 illustrates the average RMSSE distribution for one- to 15-step-ahead forecasts by our proposed FTGM and 11 benchmark models across 70 series. In these boxplots, each solid point represents the RMSSE for one series, and the black horizontal line denotes the median RMSSE per model. A lower position of the black line signifies a lower median RMSSE, and a longer box indicates greater error differences across datasets. As depicted in Fig. 3, FTGM yields both the minimum median error (1.28) and the shortest box length. The RMSSE values for FTGM are below 2 in most data series. But a few series, such as 40, 39, 20, and 60, obtain slightly higher forecasting errors (RMSSEs of 3.22, 3.10, 3.04, and 2.85) due to substantial random disturbances. To enhance predictions for these series, data preprocessing techniques like smoothing or filtering can help mitigate the influence of random disturbances. Similar results were obtained for the MASE and RMSE metrics, but we omit these plots for the sake of brevity. In summary, extensive experimental results demonstrate the superiority of FTGM over state-of-the-art forecasting methods for seasonal time series.

5.2. Ranking comparisons

Fig. 4 reports the number of times that our model and benchmark models rank in the first position and the top three concerning the RMSSE metric for one- to 15-step-ahead forecasts. As shown in Fig. 4(a), the FTGM ranks in the first position 38 times out of 70 series, while the competing models achieve the first position at most six times, demonstrating that the proposed FTGM is more effective than the benchmarks at handling seasonal time series. In Fig. 4(b), the FTGM ranks in the top three in 52 series out of 70 compared with the 11 benchmarks, meaning that the FTGM achieves better rankings with the majority of the data series. SGM and DGSM models follow closely, ranking in the top three 31 and 30 times, respectively. Furthermore, Fig. D.1 in Appendix D illustrates the ranks of the 12 models across all 70 data series for h-step-ahead forecasts. The results reveal that the FTGM obtains the best rankings across all forecast horizons.

Table 1
MASE and RMSSE: The mean and median for each model are computed on the set of MASEs and RMSSEs of all six rolling origins across all M5 datasets. For each horizon, the result for the method with the lowest error is highlighted in boldface.

MASE	$h = 1$		$h = 3$		$h = 6$		$h = 9$		$h = 12$		$h = 15$		$h = 1 - 15$	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median
FTGM	**1.09**	**0.82**	**1.24**	**1.00**	**1.30**	**1.02**	**2.04**	**1.23**	**2.02**	**1.28**	**1.99**	**1.20**	**1.65**	**1.28**
SGM	1.36	1.13	1.54	1.35	1.62	1.29	2.20	1.56	2.74	1.96	2.85	1.99	2.13	1.68
DGGM	2.95	2.65	2.75	2.72	2.33	1.95	2.76	2.00	5.90	4.55	5.58	4.64	3.28	2.98
DGSM	1.39	1.22	1.40	1.27	1.82	1.23	2.27	1.50	2.90	2.12	2.87	2.08	2.19	1.67
SNaive	2.12	1.75	2.07	1.71	2.29	2.12	2.72	2.03	3.75	3.25	3.74	3.02	2.69	2.35
SARIMA	1.29	1.14	1.60	1.48	1.91	1.60	2.64	1.93	3.47	2.54	3.62	2.56	2.44	1.98
HWES	1.17	1.03	1.57	1.45	1.91	1.74	2.76	2.29	3.51	2.79	3.59	2.48	2.47	1.97
DHR_ARIMA	1.24	1.05	1.48	1.35	1.74	1.31	2.48	1.97	3.03	2.15	3.15	2.41	2.21	1.87
BATS	1.16	0.97	1.46	1.32	1.92	1.68	2.53	2.05	3.19	2.29	3.29	2.23	2.30	1.79
TBATS	1.19	0.99	1.48	1.33	1.87	1.54	2.55	2.06	3.32	2.19	3.44	2.21	2.34	1.79
MLP	1.44	1.09	1.93	1.42	2.31	1.84	3.02	2.42	3.32	2.36	3.52	2.54	2.66	2.20
LSTM	2.29	1.79	2.28	1.96	2.49	2.06	2.88	2.50	3.13	2.56	3.25	2.62	2.73	2.35
RMSSE														
FTGM	**0.92**	**0.73**	**1.05**	**0.86**	**1.08**	**0.82**	**1.65**	**1.11**	**1.69**	**0.99**	**1.67**	**1.06**	**1.38**	**1.04**
SGM	1.31	1.09	1.42	1.25	1.49	1.23	1.92	1.40	2.38	1.81	2.46	1.86	1.87	1.46
DGGM	2.64	2.32	2.54	2.34	2.05	1.69	2.37	1.78	5.34	4.27	5.19	4.24	2.93	2.84
DGSM	1.22	1.11	1.22	1.11	1.60	1.15	1.91	1.29	2.41	1.76	2.39	1.69	1.92	1.52
SNaive	1.81	1.53	1.75	1.54	1.95	1.85	2.26	1.71	2.99	2.50	3.01	2.54	2.38	2.04
SARIMA	1.17	1.01	1.42	1.25	1.71	1.39	2.24	1.69	2.98	2.13	3.09	2.23	2.27	1.74
HWES	1.06	0.94	1.38	1.27	1.70	1.58	2.44	2.06	3.04	2.35	3.14	2.31	2.34	1.83
DHR_ARIMA	1.12	1.00	1.32	1.24	1.54	1.17	2.13	1.65	2.56	2.05	2.65	2.29	2.04	1.77
BATS	1.07	0.93	1.32	1.20	1.70	1.40	2.16	1.79	2.70	2.10	2.81	2.19	2.15	1.76
TBATS	1.09	0.93	1.33	1.26	1.67	1.38	2.20	1.70	2.88	1.99	3.01	2.20	2.23	1.75
MLP	1.39	1.01	1.76	1.28	2.11	1.54	2.65	2.10	2.90	2.15	3.04	2.13	2.52	2.00
LSTM	1.96	1.66	1.96	1.80	2.13	1.80	2.47	2.20	2.63	2.24	2.74	2.18	2.46	2.14

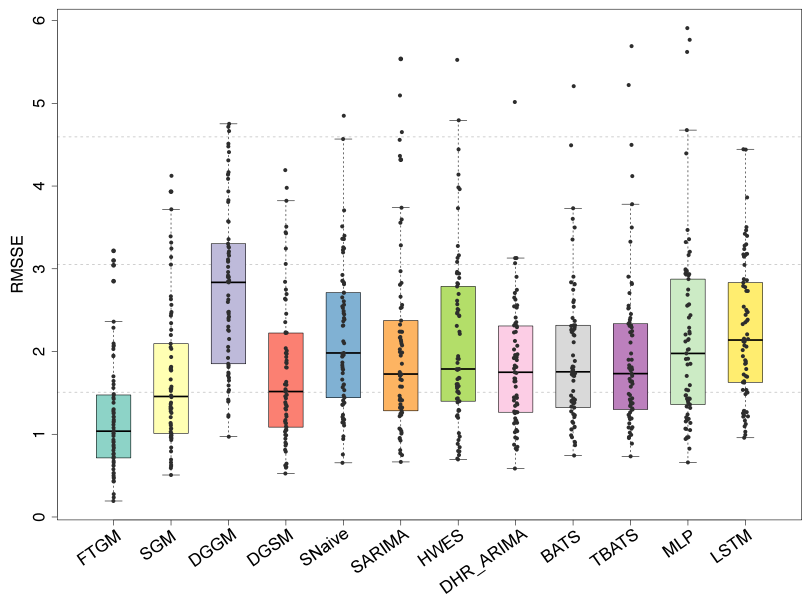

Fig. 3. RMSSE error distribution between FTGM and benchmarks for one- to 15-step-ahead forecasts.

5.3. Statistical testing analysis

To investigate statistically significant differences in the performance among different forecasting models, we conduct a multiple comparisons with the best (MCB) test (Koning, Franses, Hibon, & Stekler, 2005; Svetunkov & Petropoulos, 2018) on all M5 competition datasets. This test aims to determine whether the forecasting accuracy of different methods is significantly different from the others across multiple datasets and, if so, indicating that the model consistently outperforms its competitors. To implement this test procedure, we use the `nemenyi()` function from the `tsutils` package.

For each error measure (RMSE, MASE, and RMSSE), we calculated average ranks. Fig. 5 depicts the MCB plot of average rankings for one- to 15-step-ahead forecasts across all M5 series. Lower average ranks indicate better performance, although differences between any two methods

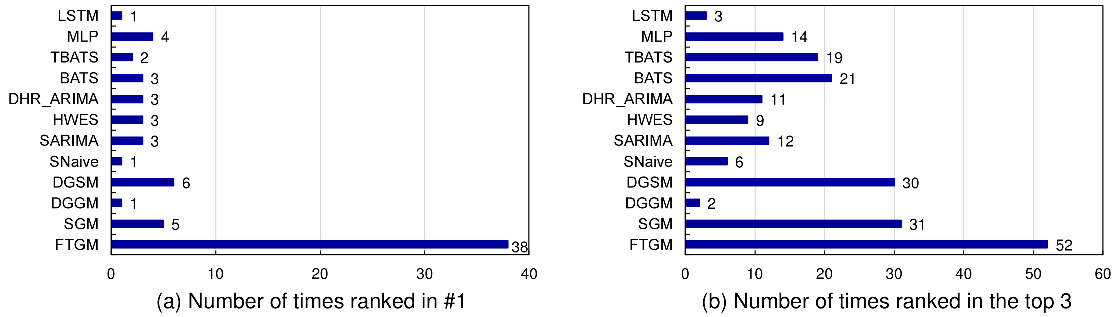

Fig. 4. Number of times that each model ranks in (a) first place and (b) top three for one- to 15-step-ahead forecasts with respect to the RMSSE.

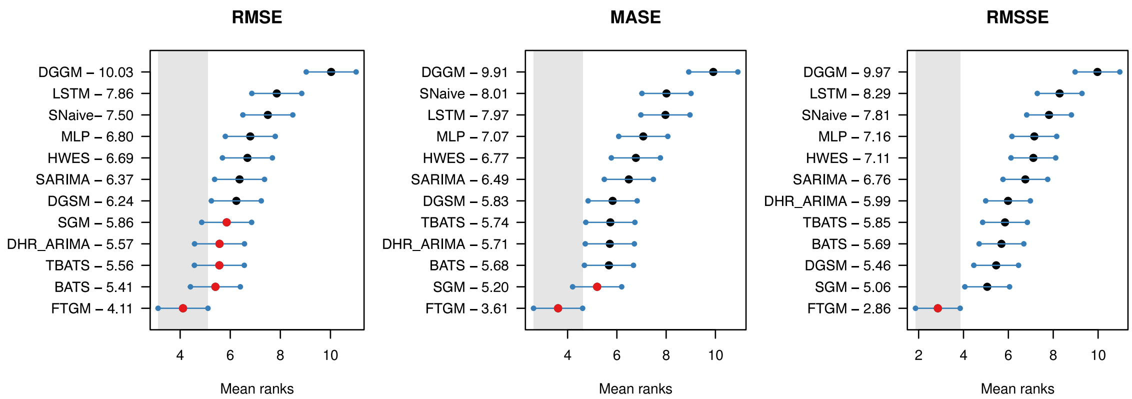

Fig. 5. Statistical significance at the 5% level: MCB test for FTGM and benchmark models for one- to 15-step-ahead forecasts across all M5 competition series with respect to three error measures.

are not considered significant if their confidence intervals overlap. Our observations reveal that the FTGM achieves the lowest average ranks in all error metrics. In terms of the RMSE, the FTGM performs significantly better than DGSM, SARIMA, HWES, MLP, SNaive, LSTM, and DGGM. Its differences are not significant from SGM, TBATS, BATS, and DHR_ARIMA, although its average ranking is better than theirs. Concerning the MASE, FTGM performs significantly better than all other models except for SGM. For the RMSSE, the FTGM demonstrates significant superiority over all benchmark methods.

Fig. 6 illustrates the MCB test results of 12 models for h-step-ahead forecast horizons across all M5 series using the RMSSE metric. Notably, the MCB results for the RMSE and MASE are in general agreement with the RMSSE. To simplify the presentation and save space, we opted to display the MCB results for the RMSSE, a widely accepted metric in the M5 Accuracy competition. As depicted in Fig. 6, the FTGM emerges as the top-performing model across various h-step-ahead forecast horizons. Specifically, for $h = 1$, the FTGM outperforms all competing models, although its differences are not statistically distinguishable from BATS, HWES, TBATS, DHR_ARIMA, SARIMA, MLP, and DGSM. At $h = 3$, FTGM significantly outperforms MLP, HWES, and SARIMA. For $h = 6$, it maintains a statistically significant advantage over BATS and TBATS. For $h = 9$, FTGM significantly outperforms other competitors, except for the seasonal grey models (SGM and DGSM). At longer forecast horizons ($h = 12$ and 15), FTGM significantly outperforms all considered competing models. It is noteworthy that BATS, TBATS, and DHR_ARIMA display no significant differences compared to the FTGM at shorter forecast horizons but significantly outperform other models. This might be attributed to the Fourier (trigonometric) functions in these models, which are well-suited for capturing seasonal variations. Moreover, seasonal grey models (SGM and DGSM) demonstrate better performance at longer horizons. This can be attributed to the nature of grey modeling as a trend model, excelling at detecting features over extended forecasting periods, thus yielding more accurate forecasts. Regarding the DGGM, it consistently performs poorly across all horizons, due to the grouping that results in the limited availability of data for modeling, with only four or five data points in each group. In contrast to traditional seasonal grey models, our FTGM is an integral matching-based model that utilizes the integral operator to substitute for the accumulative sum operator in grey models. This

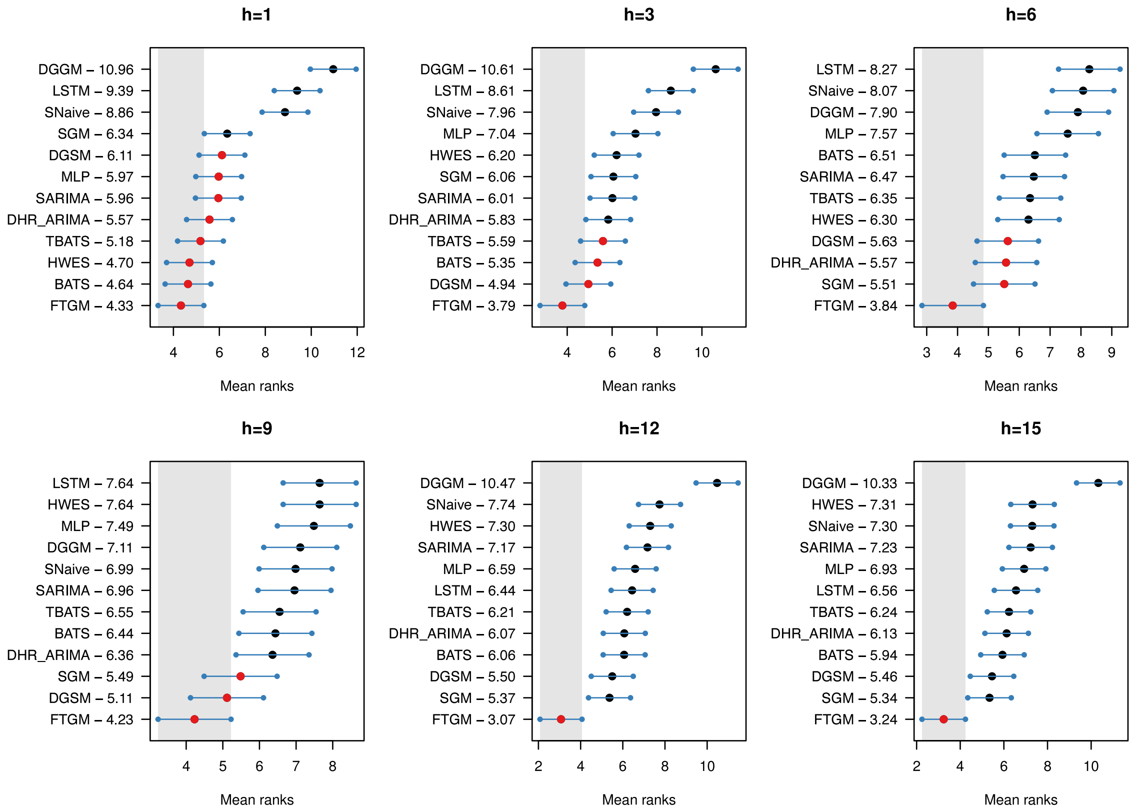

Fig. 6. Statistical significance at the 5% level: MCB test for FTGM and benchmark models for h-step-ahead forecasts across all M5 series with respect to the RMSSE.

allows our model to directly model raw data without the introduction of restoration errors. As detailed in Section 3, our approach leverages the integro-differential equation for direct forecasting, and the flexible Fourier series enhances our capacity to capture seasonal variations effectively. As a result, our model consistently produces highly accurate forecasts for seasonal time series.

5.4. Analysis of ablation experiment

To investigate whether the improvements in our proposed model are attributed to the utilization of a grey model, a Fourier grey model, a Fourier time-varying grey model, or the incorporation of the Fourier order selection process, we conduct ablation experiments on the M5 datasets. We compare our FTGM with three degenerate models: (1) a grey model that does not consider seasonality and time-varying characteristics (referred to as the GM); (2) a Fourier grey model that does not incorporate time-varying characteristics (referred to as FGM); and (3) a Fourier time-varying grey model without order selection (referred to as wFTGM, where $N = 1$). Fig. 7 reports the RMSSE comparison results of the FTGM and its degenerate models on the M5 series. The figure includes six pairwise model comparisons ((a)–(f)), each consisting of a scatter plot (top row) and a bar plot (bottom row). Specifically, the scatter plot illustrates the RMSSE comparison between the paired models on each series (each dot indicates one series). Dots above the diagonal denote that the RMSSE value for the upper-left model is higher than that of the lower-right model for these series. The bar plot depicts the average RMSSE values across all series for the two comparison models.

Comparison against the GM: The scatter plots in Fig. 7 (a)–(c) reveal that the FGM outperforms GM in the RMSSE metric across 49 out of the 70 data series, wFTGM yields improvements in 58 data series, while our FTGM achieves enhancements in 64 data series. The bar plots in Fig. 7(a)–(c) indicate that the FGM reduces the average RMSSE from 2.25 to 1.99 compared to the GM, with an average improvement of 11.55%. Meanwhile, the wFTGM achieves a 24.83% average improvement over the GM, and our FTGM yields an RMSSE of 1.38, exhibiting a remarkable enhancement of 38.84% compared to the GM. These results highlight the efficacy of employing Fourier functions to improve forecasting accuracy for grey models in seasonal time series.

Comparison against the FGM: In Fig. 7(d)–(e), the scatter plots depict that the wFTGM performs better than the FGM in the RMSSE metric across 52 out of the 70 data series, while our proposed FTGM achieves improvements in 60 data series. The bar plots show that the wFTGM enhances the FGM's average performance by 15.02% in

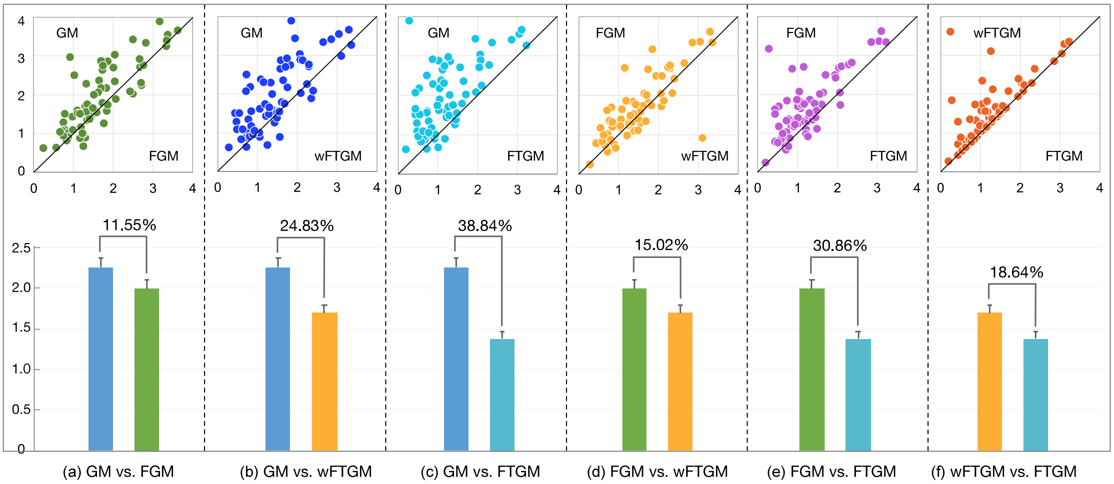

Fig. 7. Comparison of the FTGM and its degenerate models—GM, FGM, and wFTGM—for one- to 15-step-ahead forecasts on the M5 series regarding the RMSSE.

RMSSE, while FTGM exhibits a more significant improvement (30.86%) compared to the FGM. This indicates the advantage of incorporating time-varying parameters when forecasting dynamic demand series.

Comparison against the wFTGM: From Fig. 7(f), we observe that the FTGM displays improved RMSSE values in 41 out of the 70 data series and comparable results in the remaining 29 data series compared to the wFTGM. On average, our FTGM accounts for improvements by more than 18%. These results underscore the significance of the order selection approach for enhancing the model's adaptability and practicality.

Overall, the above findings from the ablation experiments show that the incorporation of Fourier functions, time-varying parameters, and the order selection approach can collectively elevate the performance of the FTGM.

5.5. Effectiveness on shorter series

To further assess the effectiveness of our proposed model at handling relatively shorter time series, we conduct experiments using subsequences extracted from the M5 data series. We follow the 'new information priority' principle from grey system theory to obtain these subsequences. Given a series $\{x(1), x(2), \ldots, x(n)\}$, we generate subsequences as $\{x(n-l+1), x(n-l+2), \ldots, x(n-h)\}$, where l denotes the sample size for training, n is the length of observations, and h represents the forecast horizon. The minimum value of l is determined based on two aspects. One is that the sample data would preferably contain two seasonal cycles. The other is that the value of l exceeds the number of parameters in the proposed model. Considering the rolling origin is six, we set $l_{min} = 27$. Since $n = 63$ and $h = 15$ in our experiments, we get $l_{max} = 48$. Thus, the range of l varies from 27 to 48.

Table 2 reports the average RMSSE values for 12 comparison models, covering sample sizes ranging from 27 to 48. It is obvious that, in general, the RMSSE increases as the sample size decreases for the majority of the comparison models. This can be attributed to the fact that smaller sample sizes tend to result in less clear or stable seasonal patterns. Notably, the FTGM consistently outperforms all others, yielding the lowest RMSSE values across various sample sizes, followed closely by the SGM, DHR-ARIMA, and DGSM. Even in the case of the smallest sample, our model stands out, with average RMSSE values not exceeding 2 (ranging from 1.81 to 1.38). These results highlight the effectiveness of our proposed model at handling small sample data. On the other hand, the standard deviations of the RMSSE in the FTGM are significantly smaller than those of other models across different sample lengths, illustrating the robustness of the FTGM. Additionally, we observe that some models, such as the DGGM and LSTM, struggle to make forecasts with small sample data. The DGGM groups the same monthly data, resulting in sample sizes that are too small to train within each group when the subsequence is short. That is, seasonal time series forecasting is influenced not only by the sample size but also by the seasonal cycles. For LSTM, it may not have enough data to estimate the parameters in very small samples. As demonstrated, the proposed FTGM emerges as an accurate, robust, and practical approach to seasonal demand forecasting with small sample data.

6. Conclusions and future research

In this paper, we proposed a Fourier time-varying grey model as a flexible time series forecasting method for seasonal demand in retail. The FTGM leverages Fourier functions to approximate the time-varying parameters, enabling it to model seasonal variations effectively. Unlike the parameter estimation method used in traditional

Table 2
Average RMSSE by 12 models with six rolling origins across different sample sizes of the M5 datasets. For each series size, the result for the method with the lowest error is highlighted in boldface.

Size	FTGM	SGM	DGGM	DGSM	SNaive	SARIMA	HWES	DHR–ARIMA	BATS	TBATS	MLP	LSTM
27	$\mathbf{1.81}_{\pm 2.13}$	$2.48_{\pm 2.04}$	–	$2.71_{\pm 2.12}$	$2.43_{\pm 1.98}$	$2.98_{\pm 2.54}$	$3.01_{\pm 2.25}$	$2.37_{\pm 1.93}$	$2.82_{\pm 2.59}$	$4.97_{\pm 18.28}$	$3.37_{\pm 4.85}$	–
28	$\mathbf{1.78}_{\pm 2.41}$	$2.37_{\pm 2.02}$	–	$2.63_{\pm 2.08}$	$2.38_{\pm 1.95}$	$2.85_{\pm 2.34}$	$2.92_{\pm 2.20}$	$2.33_{\pm 1.95}$	$2.69_{\pm 2.47}$	$3.85_{\pm 10.41}$	$3.22_{\pm 4.27}$	–
29	$\mathbf{1.60}_{\pm 0.98}$	$2.28_{\pm 2.02}$	–	$2.54_{\pm 2.06}$	$2.34_{\pm 1.95}$	$2.70_{\pm 2.12}$	$2.91_{\pm 2.18}$	$2.22_{\pm 1.89}$	$2.62_{\pm 2.45}$	$5.14_{\pm 20.93}$	$3.14_{\pm 4.21}$	–
30	$\mathbf{1.67}_{\pm 1.02}$	$2.22_{\pm 2.03}$	–	$2.47_{\pm 2.05}$	$2.33_{\pm 1.95}$	$2.67_{\pm 2.13}$	$2.61_{\pm 2.08}$	$2.20_{\pm 1.90}$	$4.73_{\pm 17.5}$	$4.92_{\pm 18.31}$	$2.70_{\pm 2.50}$	–
31	$\mathbf{1.69}_{\pm 1.38}$	$2.18_{\pm 2.06}$	–	$2.33_{\pm 2.05}$	$2.32_{\pm 1.95}$	$2.82_{\pm 2.27}$	$2.47_{\pm 2.02}$	$2.20_{\pm 1.91}$	$2.61_{\pm 2.75}$	$2.80_{\pm 2.69}$	$2.73_{\pm 2.54}$	–
32	$\mathbf{1.44}_{\pm 1.32}$	$2.17_{\pm 2.08}$	–	$2.26_{\pm 2.07}$	$2.32_{\pm 1.95}$	$2.80_{\pm 2.54}$	$2.43_{\pm 2.01}$	$2.14_{\pm 1.92}$	$2.67_{\pm 2.83}$	$2.81_{\pm 2.88}$	$2.79_{\pm 2.75}$	–
33	$\mathbf{1.64}_{\pm 1.56}$	$2.16_{\pm 2.08}$	–	$2.22_{\pm 2.09}$	$2.30_{\pm 1.94}$	$2.95_{\pm 2.49}$	$2.45_{\pm 2.05}$	$2.10_{\pm 1.91}$	$2.51_{\pm 2.39}$	$2.70_{\pm 2.76}$	$2.57_{\pm 2.21}$	$3.64_{\pm 2.15}$
34	$\mathbf{1.50}_{\pm 1.69}$	$2.16_{\pm 2.10}$	–	$2.20_{\pm 2.11}$	$2.30_{\pm 1.94}$	$2.79_{\pm 2.42}$	$2.35_{\pm 2.02}$	$2.09_{\pm 1.91}$	$2.51_{\pm 2.40}$	$2.61_{\pm 2.43}$	$2.60_{\pm 2.25}$	$3.65_{\pm 2.16}$
35	$\mathbf{1.45}_{\pm 1.68}$	$2.15_{\pm 2.08}$	–	$2.18_{\pm 2.11}$	$2.30_{\pm 1.94}$	$2.96_{\pm 2.72}$	$2.32_{\pm 2.01}$	$2.07_{\pm 1.91}$	$2.49_{\pm 2.27}$	$2.54_{\pm 2.33}$	$2.65_{\pm 2.51}$	$3.69_{\pm 2.18}$
36	$\mathbf{1.45}_{\pm 1.72}$	$2.12_{\pm 2.06}$	–	$2.15_{\pm 2.09}$	$2.29_{\pm 1.93}$	$2.83_{\pm 2.65}$	$2.33_{\pm 2.01}$	$2.06_{\pm 1.89}$	$2.44_{\pm 2.28}$	$2.57_{\pm 2.32}$	$2.48_{\pm 2.06}$	$3.73_{\pm 2.20}$
37	$\mathbf{1.38}_{\pm 1.77}$	$2.08_{\pm 2.07}$	–	$2.12_{\pm 2.07}$	$2.29_{\pm 1.92}$	$2.68_{\pm 2.37}$	$2.44_{\pm 2.05}$	$2.03_{\pm 1.89}$	$2.42_{\pm 2.26}$	$2.43_{\pm 2.24}$	$2.49_{\pm 2.06}$	$3.78_{\pm 2.22}$
38	$\mathbf{1.52}_{\pm 1.81}$	$2.04_{\pm 2.05}$	–	$2.09_{\pm 2.08}$	$2.28_{\pm 1.93}$	$2.65_{\pm 2.31}$	$2.45_{\pm 2.12}$	$2.03_{\pm 1.89}$	$2.39_{\pm 2.24}$	$2.36_{\pm 2.20}$	$2.48_{\pm 2.20}$	$3.82_{\pm 2.26}$
39	$\mathbf{1.50}_{\pm 1.81}$	$1.99_{\pm 2.02}$	–	$2.06_{\pm 2.06}$	$2.29_{\pm 1.94}$	$2.57_{\pm 2.32}$	$2.47_{\pm 2.14}$	$2.03_{\pm 1.90}$	$2.32_{\pm 2.20}$	$2.35_{\pm 2.17}$	$2.49_{\pm 2.28}$	$2.76_{\pm 1.88}$
40	$\mathbf{1.42}_{\pm 1.78}$	$1.95_{\pm 2.01}$	–	$2.03_{\pm 2.04}$	$2.30_{\pm 1.93}$	$2.59_{\pm 2.32}$	$2.46_{\pm 2.09}$	$2.01_{\pm 1.89}$	$2.29_{\pm 2.11}$	$2.46_{\pm 2.60}$	$2.66_{\pm 2.76}$	$2.76_{\pm 1.89}$
41	$\mathbf{1.49}_{\pm 1.78}$	$1.89_{\pm 2.02}$	$3.68_{\pm 2.25}$	$2.01_{\pm 2.04}$	$2.30_{\pm 1.94}$	$2.29_{\pm 2.08}$	$2.35_{\pm 2.03}$	$2.02_{\pm 1.90}$	$2.31_{\pm 2.12}$	$2.36_{\pm 2.21}$	$2.48_{\pm 2.21}$	$2.72_{\pm 1.89}$
42	$\mathbf{1.42}_{\pm 1.80}$	$1.88_{\pm 2.04}$	$3.59_{\pm 2.17}$	$1.99_{\pm 2.05}$	$2.30_{\pm 1.93}$	$2.28_{\pm 2.09}$	$2.40_{\pm 2.22}$	$1.99_{\pm 1.88}$	$2.34_{\pm 2.23}$	$2.37_{\pm 2.38}$	$2.55_{\pm 2.17}$	$2.69_{\pm 1.89}$
43	$\mathbf{1.50}_{\pm 1.79}$	$1.89_{\pm 2.05}$	$3.51_{\pm 2.12}$	$1.97_{\pm 2.06}$	$2.31_{\pm 1.93}$	$2.27_{\pm 2.10}$	$2.41_{\pm 2.14}$	$1.98_{\pm 1.88}$	$2.34_{\pm 2.18}$	$2.46_{\pm 2.67}$	$2.51_{\pm 2.15}$	$2.69_{\pm 1.89}$
44	$\mathbf{1.47}_{\pm 1.52}$	$1.91_{\pm 2.09}$	$3.45_{\pm 2.10}$	$1.97_{\pm 2.07}$	$2.32_{\pm 1.93}$	$2.29_{\pm 2.09}$	$2.41_{\pm 2.17}$	$2.01_{\pm 1.91}$	$2.41_{\pm 2.30}$	$2.40_{\pm 2.64}$	$2.48_{\pm 2.09}$	$2.70_{\pm 1.88}$
45	$\mathbf{1.54}_{\pm 1.78}$	$1.94_{\pm 2.11}$	$3.40_{\pm 2.09}$	$1.98_{\pm 2.10}$	$2.33_{\pm 1.94}$	$2.19_{\pm 2.01}$	$2.47_{\pm 2.24}$	$2.00_{\pm 1.92}$	$2.37_{\pm 2.34}$	$2.40_{\pm 2.77}$	$2.54_{\pm 2.40}$	$2.42_{\pm 1.90}$
46	$\mathbf{1.52}_{\pm 1.77}$	$1.97_{\pm 2.12}$	$3.37_{\pm 2.10}$	$2.00_{\pm 2.12}$	$2.35_{\pm 1.95}$	$2.21_{\pm 2.05}$	$2.40_{\pm 2.11}$	$2.01_{\pm 1.92}$	$2.43_{\pm 2.40}$	$2.48_{\pm 2.64}$	$2.45_{\pm 2.11}$	$2.42_{\pm 1.80}$
47	$\mathbf{1.45}_{\pm 1.78}$	$1.99_{\pm 2.14}$	$3.35_{\pm 2.11}$	$2.02_{\pm 2.12}$	$2.36_{\pm 1.96}$	$2.21_{\pm 2.03}$	$2.43_{\pm 2.27}$	$2.03_{\pm 1.95}$	$2.27_{\pm 2.17}$	$2.35_{\pm 2.52}$	$2.50_{\pm 2.25}$	$2.45_{\pm 1.89}$
48	$\mathbf{1.38}_{\pm 1.69}$	$2.00_{\pm 2.14}$	$3.33_{\pm 2.12}$	$2.03_{\pm 2.14}$	$2.38_{\pm 1.97}$	$2.27_{\pm 2.11}$	$2.34_{\pm 2.09}$	$2.05_{\pm 1.94}$	$2.15_{\pm 1.98}$	$2.23_{\pm 2.12}$	$2.52_{\pm 2.38}$	$2.46_{\pm 1.90}$

Note: '–' indicates that the model is unable to make forecasts for the current series size.

grey models, our model uses an integral matching approach with integral operators and least squares to estimate the model parameters. Furthermore, we developed a data-driven order selection method, combined with the best subset selection and a validation set, to adaptively determine the appropriate order for the model.

We assessed the predictive performance of our proposed FTGM using real-world data from the M5 competition, composed of 70 monthly product demand time series from seven departments and 10 stores. We compared the proposed model against state-of-the-art forecasting methods renowned for their proficiency at handling seasonal data. These methods included seasonal grey forecasting models (SGM, DGGM, and DGSM), statistical forecasting methods (SNaive, SARIMA, HWES, DH-ARIMA, BATS, and TBATS), and neural networks (MLP and LSTM). We employed the RMSE, MASE, and RMSSE metrics to evaluate the forecasting accuracy. Our results showed that the proposed FTGM consistently outperformed benchmark methods across the entire dataset under all considered error metrics. In particular, it was outstandingly superior to all competing models over a long forecast horizon. Moreover, our ablation studies and robustness testing on shorter series substantiated the effectiveness of the FTGM, marking it as a valuable contribution to the field of seasonal retail forecasts, with potential utility across various practical applications.

Our current study also creates many potentially fruitful opportunities for further research. One of the potential strategies for the estimation of high-dimensional models on small samples is the shrinkage of parameters, which could be investigated in subsequent research papers. For model selection strategies, while we currently determine the appropriate Fourier order, the selection of the FTGM's structure remains open. Further investigation into the model's structure through sparse estimation is a viable direction. In practical retail, data collection is often more granular. Therefore, there is an opportunity to delve into intermittent demand forecasting at the SKU level, particularly at higher temporal frequencies (e.g., daily or weekly data). Additionally, it is worth exploring the impact of multiple factors on product demand, such as price and promotions. Future work can also look at preprocessing techniques, including data smoothing and filtering methods, to reduce the influence of noise.

CRediT authorship contribution statement

Lili Ye: Conceptualization, Methodology, Validation, Writing – original draft. **Naiming Xie:** Supervision, Funding acquisition. **John E. Boylan:** Writing – review & editing. **Zhongju Shang:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and codes are available at https://github.com/yll-66/ftgmmodel.

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Appendix A. Derivation of the integral FTGM

Here, we show the derivation of the integral equation (8) from the integro-differential equation (5).

By integrating both sides of Eq. (5), we have:

$$\int_{t_1}^{t} dx(\tau) = \int_{t_1}^{t} a(\tau)x(\tau)d\tau + \int_{t_1}^{t} a'(\tau)\left(\eta_y + \int_{t_1}^{\tau} x(s)ds\right)d\tau + \int_{t_1}^{t} b'(\tau)d\tau \tag{A.1}$$

Based on the Newton–Leibniz formula, substituting Eq. (4) into Eq. (A.1) gives:

$$x(t) - x(t_1) = \int_{t_1}^{t} a(\tau)x(\tau)d\tau + \int_{t_1}^{t} a'(\tau)y(\tau)d\tau + b(t) - b(t_1) \tag{A.2}$$

Differentiating both sides of Eq. (4) with respect to t, we have that:

$$\frac{d}{dt}y(t) = \frac{d}{dt}\left(\eta_y + \int_{t_1}^{t} x(s)ds\right) = x(t) \tag{A.3}$$

Integrating the first term of the right-hand side of Eq. (A.2) by parts, we have:

$$\int_{t_1}^{t} a(\tau)x(\tau)d\tau = \int_{t_1}^{t} a(\tau)dy(\tau) = a(t)y(t) - \int_{t_1}^{t} a'(\tau)y(\tau)d\tau - a(t_1)\eta_y \tag{A.4}$$

Substituting Eq. (A.4) into Eq. (A.2) gives:

$$x(t) = a(t)y(t) + b(t) - a(t_1)\eta_y - b(t_1) + x(t_1) \tag{A.5}$$

Substituting the initial value $x(t_1) = a(t_1)\eta_y + b(t_1)$ into Eq. (A.5), one gets:

$$x(t) = a(t)y(t) + b(t),\ \ t \geq t_1, \tag{A.6}$$

Without loss of generality, we let the initial value be $x(t_1) = \eta_x = \eta_y = \frac{b(t_1)}{1-a(t_1)}$. Then, substituting Eq. (4) into Eq. (A.6), we have the integral equation:

$$x(t) = a(t)\left(x(t_1) + \int_{t_1}^{t} x(s)ds\right) + b(t),\ \ t \geq t_1, \tag{A.7}$$

Appendix B. Name of aggregated time series from M5 competition data

- Series 1–7: Hobbies_1_CA_1, Hobbies_2_CA_1, Household_1_CA_1, Household_2_CA_1, Foods_1_CA_1, Foods_2_CA_1, Foods_3_CA_1;
- Series 8–14: Hobbies_1_CA_2, Hobbies_2_CA_2, Household_1_CA_2, Household_2_CA_2, Foods_1_CA_2, Foods_2_CA_2, Foods_3_CA_2;
- Series 15–21: Hobbies_1_CA_3, Hobbies_2_CA_3, Household_1_CA_3, Household_2_CA_3, Foods_1_CA_3, Foods_2_CA_3, Foods_3_CA_3;
- Series 22–28: Hobbies_1_CA_4, Hobbies_2_CA_4, Household_1_CA_4, Household_2_CA_4, Foods_1_CA_4, Foods_2_CA_4, Foods_3_CA_4;
- Series 29–35: Hobbies_1_TX_1, Hobbies_2_TX_1, Household_1_TX_1, Household_2_TX_1, Foods_1_TX_1, Foods_2_TX_1, Foods_3_TX_1;
- Series 36–42: Hobbies_1_TX_2, Hobbies_2_TX_2, Household_1_TX_2, Household_2_TX_2, Foods_1_TX_2, Foods_2_TX_2, Foods_3_TX_2;
- Series 43–49: Hobbies_1_TX_3, Hobbies_2_TX_3, Household_1_TX_3, Household_2_TX_3, Foods_1_TX_3, Foods_2_TX_3, Foods_3_TX_3;
- Series 50–56: Hobbies_1_WI_1, Hobbies_2_WI_1, Household_1_WI_1, Household_2_WI_1, Foods_1_WI_1, Foods_2_WI_1, Foods_3_WI_1;
- Series 57–63: Hobbies_1_WI_2, Hobbies_2_WI_2, Household_1_WI_2, Household_2_WI_2, Foods_1_WI_2, Foods_2_WI_2, Foods_3_WI_2;

Table C.1
An example: Parameter estimations of our FTGM and benchmarks for Hobbies_1_CA_1 series.

Model	Model form	Parameter estimations
FTGM	FTGM(1,1,N) with $N = 1$	$\hat{\boldsymbol{\theta}} = [0.0056, 10771.06, 0.0013, -0.0023, -826.69, 1135.56]^{\top}$
SGM	SGM(1,1)	$\hat{\boldsymbol{\theta}} = [-0.0049, 11043.6753]^{\top}$, $f_s = [0.9570, 1.0159, 1.0563, 1.0063, 1.0864, 1.0374, 1.0261, 0.9171, 1.0066, 0.9015, 0.970, 0.958]^{\top}$
DGGM	DGGM(1,1)	$\hat{\boldsymbol{\theta}}_1 = [-0.3883, 2970.6]^{\top}, \hat{\boldsymbol{\theta}}_2 = [-0.0564, 10133.9]^{\top}, \hat{\boldsymbol{\theta}}_3 = [-0.0360, 11562.8]^{\top}$, $\hat{\boldsymbol{\theta}}_4 = [-0.0335, 12064]^{\top}, \hat{\boldsymbol{\theta}}_5 = [-0.1228, 8779.8]^{\top}, \hat{\boldsymbol{\theta}}_6 = [-0.0410, 12431.4]^{\top}$, $\hat{\boldsymbol{\theta}}_7 = [-0.0993, 9829]^{\top}, \hat{\boldsymbol{\theta}}_8 = [-0.1688, 7606.1]^{\top}, \hat{\boldsymbol{\theta}}_9 = [-0.3143, 3759.1]^{\top}$, $\hat{\boldsymbol{\theta}}_{10} = [-0.4454, 1421.4]^{\top}, \hat{\boldsymbol{\theta}}_{11} = [-0.5542, -290.5]^{\top}, \hat{\boldsymbol{\theta}}_{12} = [-0.6074, -1760.2]^{\top}$
DGSM	DGSM(1,1)	$\hat{\boldsymbol{\theta}} = [1.0050, 10361.39, 11402.00, 11837.86, 11156.75, 12083.19, 11412.24, 11209.03, 10180.45, 11228.02, 9871.12, 10660.64, 10448.63]^{\top}$
SARIMA	ARIMA(0,1,0)(1,1,0) [12]	sma1 $= -0.7310$
HWES	ETS(A,A,A)	$\alpha = 0.7668$, $\beta = 0.0002$, $\gamma = 0.0091$
DH_ARIMA	ARIMA(1,1,0) with $K = 3$	ar1 $= -0.3727$, S1_12 $= 410.3778$, C1_12 $= -415.3910$, S2_12 $= -78.3911$, C2_12 $= -32.0380$, S3_12 $= 26.6789$, C3_12 $= -138.4705$
BATS	BATS(0.262, {3, 1}, −, {12})	$\lambda = 0.262$, $\alpha = 0.07$, $\gamma = -0.041$, ar1 $= 0.0629$, ar2 $= 0.4675$, ar3 $= 0.43$, ma1 $= 0.5037$
TBATS	TBATS(0.26, {5, 0}, 1, {< 12, 5 >})	$\lambda = 0.26$, $\alpha = 0.344$, $\beta = -0.0073$, $\gamma_1 = -0.000378$, $\gamma_2 = 0.000064$, ar1 $= -0.1415$, ar2 $= 0.3929$, ar3 $= 0.1401$, ar4 $= 0.12$, ar5 $= -0.2214$
MLP	MLP(reps=20, hd=7)	repetitions=20, hidden nodes=7, lags=(5,11), difforder=1
LSTM	LSTM(24, input_shape = (timesteps, 1))	timesteps = 12, hidden neurons = 24, activation = ‘tanh’, batch size = 6, loss = ‘RMSE’, optimizer = ‘adam’, epochs = 20

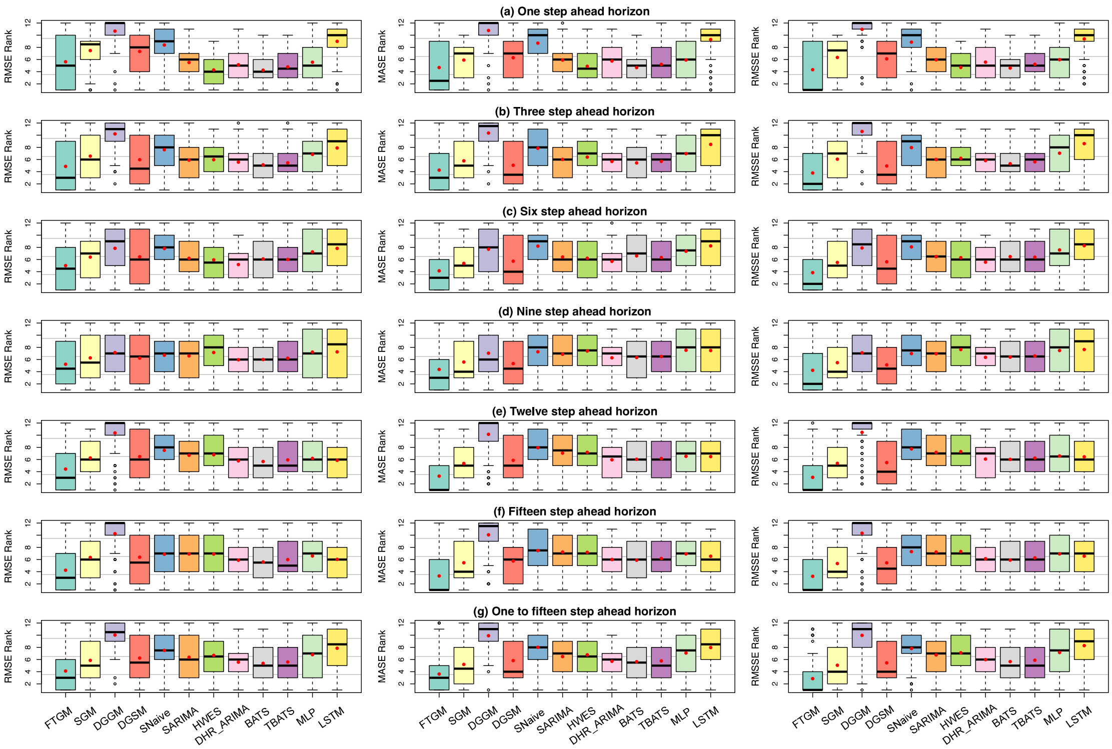

Fig. D.1. RMSE, MASE, and RMSSE ranks of 12 models across all 70 department-store M5 series for h-step-ahead forecast horizons. The box is drawn from 25% to 75% with two whiskers representing an equal 1:5 interquartile range from the edges of the box. The line inside the box is the median, and the red points indicate the mean in all datasets.

- Series 64–70: Hobbies_1_WI_3, Hobbies_2_WI_3, Household_1_WI_3, Household_2_WI_3, Foods_1_WI_3, Foods_2_WI_3, Foods_3_WI_3;

Appendix C. Supplementary data

See Table C.1.

Appendix D. Figure of error ranks

See Fig. D.1.

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